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ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION

TECHNOLOGY

ELEMENT: DESIGN COMPUTER NETWORK

UNIT CODE: IT/CU/ICTA/CR/02/5/MA

UNIT NAME: NETWORK DESIGN & MANAGEMENT

LEVEL 5, MODULE 3

TERM 3

TOTAL MARKS: 50 MARKS

Name... Joy Nkatha ..... Date.....

Sign..... Cdacc regNo.....

**Instructions:**

*Answer all questions.*

1. Define a computer network and state two benefits of computer networking. (4 marks)
2. State and explain four types of computer networks based on geographical coverage. (8 marks)
3. Explain the following network topologies:
  - a) Star
  - b) Ring
  - c) Mesh
  - d) Hybrid
  - e) Point-to-point (10 marks)
4. State four components of a computer network and explain the function of each. (8 marks)
5. State three examples of network transmission media and give one characteristic of each. (6 marks)
6. Explain the following stages involved in collecting and handling computer network user requirements:
  - a) User requirements identification

a group of interconnected computers and devices that communicate and share data.

- (i) R. Users can share resources.
- (ii) Users can communicate easily.

2. LAN - cover a small geographical area.

MAN - cover a large town, connecting several LANs.

WAN - cover a large geographical area.

PAN - cover a small area around an individual.

3. Star - All computers are connected to a central device.

(b) Ring - Each device is connected to two other devices forming a ring.

(c) Mesh - Devices are connected to other devices providing multiple paths.

(d) Hybrid - A network formed by combining two or more different topologies.

(e) Point-to-point - A direct connection between devices.

4. Network Interface card - allow computer to connect to a network.

Router - Connect networks and direct data between them.

Switch - connects multiple devices on a LAN.

5. Twisted pair cable - commonly used in LANs.

Fibre-optic cable - provide high speed data transmission.

Radio waves - transmit data without physical cable.

6. Determine what users need from the network.

6. Examine the collected requirements to determine whether they are necessary.

7. Record the requirements in a document that can be used in designing the network.

7. ~~to~~ Physical

Describes the ~~actual~~ arrangement of the network

Logical

Describes how data flows and how the network is organized.

Includes cables, switches, routers.

Includes IP addresses, subnets, protocols and routing.

8. Greenfield network site is a new site with no existing ~~network~~ infrastructure while Brownfield network site is a site that has existing network infrastructure.

9. Provide reliable connectivity and communication.  
Ensure efficient and secure sharing of resources?

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